UNITARES: Information-Theoretic Governance of Heterogeneous Agent Fleets

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Abstract

We present UNITARES, a governance framework for heterogeneous fleets of autonomous AI agents. Each agent carries a four-dimensional state vector $\mathbf{x} = (E, I, S, V)^\top$ intended to support continuous governance rather than discrete after-the-fact filtering. The central design constraint is heterogeneity: the populations we govern include embodied agents with sensor-driven state, persistent autonomous services, session-bounded coding assistants, and ephemeral parser agents. These do not share an output modality, a tempo, or a healthy operating point, so fleet-wide normalization is the wrong target.

UNITARES addresses this with two structural moves. First, it reinterprets the EISV coordinates in information-theoretic terms: S as response-distribution entropy, I as context-response mutual information, E as negative variational free energy or a resource-rate proxy, and V as an accumulated free-energy residual. Second, it makes normalization class-conditional: scale constants and healthy operating points are functions of an agent class keyed on existing identity tags, with fleet-wide defaults retained only as fallback behavior.

Because UNITARES was already deployed, regrounding the coordinates raised a live-systems problem in addition to a mathematical one. We therefore introduce a pipeline-ordering migration mechanism that preserves governance behavior while allowing grounded values to populate canonical response fields. Stability of the dynamics is preserved under the reformulation and established by contraction analysis (Appendix B).

We illustrate the framework with a production deployment snapshot from February 20, 2026, covering 903 registered agents and 75 active agents. These results should be read as partial deployment evidence for the framework and migration strategy, not as full empirical validation of the class-conditional grounding. Per-class calibration results and higher-tier estimators (log-prob and multi-sample) are deferred to subsequent work.

Keywords: autonomous governance, information theory, variational free energy, heterogeneous multi-agent systems, class-conditional calibration, AI safety

1 Introduction

Autonomous AI agents operating in production environments face a fundamental tension: they must explore to learn, yet remain safe while doing so. Existing approaches to agent governance fall into two camps. *Hard constraints* (guardrails, output filters) provide safety but impede capability. *Soft monitoring* (logging, dashboards) preserves capability but offers no formal guarantees.

We argue that governance should be *intrinsic*—a continuous dynamical process that shapes agent behavior from within, much as homeostatic regulation governs biological organisms. This paper presents UNITARES (**Uni**fied **T**hermodynamic **A**daptive **Re**gulation **S**ystem), a framework in which each agent carries a four-

dimensional state $\mathbf{x} = (E, I, S, V)^\top$ whose evolution is governed by coupled differential equations. The state provides a continuous, interpretable signal for governance decisions—when to proceed, when to pause, when to escalate.

The framework's earliest formulations Wang [2026a,b] were thermodynamically inspired but phenomenologically constructed: the coordinates were named after physical quantities that they did not actually compute. This paper re-grounds the coordinates information-theoretically while preserving the dynamics, the verdicts, and the production deployment. We further address a design constraint that earlier formulations did not surface: the populations the framework governs are heterogeneous. They differ in tempo by four orders of magnitude, in output modality (text, sensor, system action), and in healthy operating point. A governance framework that collapses this population to a single state distribution—what we call *cosmological soup*—loses signal exactly where signal matters. The contributions of this paper are organized around these two reframings.

1.1 Contributions

This paper makes five contributions:

1. **Framework reframing for heterogeneous fleets** (Section 2): We identify heterogeneity, rather than single-population stability, as the governing design constraint for a production agent-governance framework.

2. **Information-theoretic reinterpretation of EISV** (Section 4): We replace the earlier thermodynamic labeling with computable information-theoretic semantics for the EISV coordinates, together with tiered estimation recipes and provenance tags.

3. **Class-conditional calibration protocol** (Section 9): We formalize scale constants and healthy operating points as class-indexed quantities derived from identity structure, with explicit fallback behavior for sparse or unclassified classes.

4. **Behavior-preserving migration mechanism for deployed systems** (Section 10): We introduce a pipeline-ordering strategy that separates decision-producing and presentation-producing stages so semantic re-grounding can occur without changing live governance behavior during migration.

5. **Partial deployment evidence** (Section 13): We report a production snapshot of the deployed system to characterize operating regimes, saturation behavior, and observed fleet dynamics. These results are descriptive and migration-oriented; full class-conditional empirical validation is left to subsequent work.

1.2 Limitations and Scope

This paper is a framework-and-migration paper, not a completed empirical validation paper. Its primary claims are architectural and methodological: that heterogeneous fleets require class-conditional calibration, that the EISV coordinates can be reinterpreted information-theoretically without changing the governing dynamics, and that a deployed governance system can be re-grounded by separating decision-producing from presentation-producing stages in the request pipeline.

Accordingly, three boundaries matter. First, the class-conditional calibration protocol is specified here, but the per-class empirical constants are not yet reported. Second, the highest-fidelity estimators for grounded quantities (logprob-based and multi-sample computation) remain future instrumentation work; the current deployment relies on degraded-mode fallbacks where those signals are unavailable. Third, the deployment results in Section 13 are presented as a single descriptive production snapshot rather than a controlled validation study across stable parameter regimes.

The paper should therefore be read as establishing the framework, the migration mechanism, and the shape of the empirical program, while deferring full class-conditional validation to subsequent work.

1.3 Related Work

Information-Theoretic Monitoring. Shannon entropy and mutual information are standard tools for characterizing model output distributions; calibration analysis Guo et al. [2017] treats the related question of confidence-outcome alignment. The variational free-energy principle Friston [2010] provides a unifying frame in which agent surprise corresponds to free energy and adaptive agents minimize expected surprise. UNITARES adapts these information-theoretic quantities to the governance setting and treats the choice of computation tier (logprob, multi-sample, heuristic) as a deployment configuration rather than an assumption.

Contraction Theory. Lohmiller and Slotine Lohmiller and Slotine [1998] established contraction analysis as a tool for nonlinear stability, later extended to Riemannian metrics Slotine [2003]. We adapt this framework to the governance domain, where the "plant" is an agent's state vector rather than a physical system. Contraction is preserved by the grounded reformulation but is no longer the headline contribution; see Appendix B.

AI Safety and Governance. Constitutional AI Bai et al. [2022] and RLHF Ouyang et al. [2022] govern model outputs through training objectives, while output filters and guardrails Rebedea et al. [2023] enforce constraints at inference time. These approaches treat safety as a discrete property (safe/unsafe) applied at the boundary of agent action. UNITARES instead models safety as a continuous dynamical state that evolves between actions, providing leading indicators of degradation rather than binary verdicts after the fact.

Ravindran Ravindran [2025] proposes a Moral Anchor System (MAS) that detects value drift using a three-dimensional state vector with Bayesian inference and LSTM-based forecasting. While MAS and UNITARES share the goal of detecting ethical drift, they differ fundamentally: MAS is a statistical detection pipeline that predicts drift from observed behavior, whereas UNITARES models drift as a dynamical force that couples into the state equations, providing contraction-theoretic stability guarantees that statistical methods cannot offer.

Control Barrier Functions (CBFs) Ames et al. [2019] enforce forward invariance of safe sets for multi-agent systems. CBFs guarantee that the system remains within a safe region but do not provide convergence to a desired operating point. Contraction theory, by contrast, guarantees exponential convergence to a unique trajectory, a strictly stronger property that subsumes set invariance when the equilibrium lies within the safe set.

Adaptive Control. PID control is classical Åström and Murray [2008]; our contribution is applying it to governance thresholds with phase-aware reference tracking, connecting it to the EISV state vector through the coherence function.

Calibration. Confidence calibration has a rich literature in forecasting and ML Guo et al. [2017], Niculescu-Mizil and Caruana [2005]. Our ethical drift vector uses calibration deviation as one of four measurable components, connecting calibration to governance dynamics.

2 Heterogeneous Agent Fleets

A central design constraint of UNITARES is that the population it governs is heterogeneous. The deployments that motivated the framework include an embodied creature with I^2C sensors sampled at sub-Hertz rates, a cron-driven janitorial agent that wakes once every 30 minutes, an analytical observer that streams continuously over a websocket, a session-bounded coding assistant generating text at 40–80 tokens per second, and ephemeral parser agents whose entire lifecycle measures in milliseconds. These do not share a tempo. Their outputs are not in the same units. They do not have a common healthy operating point. A governance framework that collapses them to a single state distribution loses signal exactly where signal matters.

2.1 The Homogenization Failure Mode

Most agent-monitoring frameworks assume a single inference layer producing comparable outputs at comparable rates—typically a single LLM behind a single client API. Under that assumption, a fleet-wide normalization is harmless: every agent's response distribution lives on the same support; every agent's expected throughput sits in the same envelope; a single percentile of a single quantity describes the whole population.

UNITARES violates each of these assumptions by construction. The framework runs governance check-ins from agents whose outputs are not text (sensor readings; system actions; drawing strokes), whose tempos differ by four orders of magnitude (continuous streams to days-between-wakes), and whose "healthy operating point" is genuinely class-dependent. A divergent/exploratory agent operating at high entropy is doing exactly what we want; flagging it as incoherent against an analytical-class baseline is not a governance signal but a category error.

We refer to the failure mode of fleet-wide normalization as *cosmological soup*: a state distribution that is maximally entropic, with no class structure and no distinguishability between populations whose dynamics are genuinely different. The information-theoretic grounding presented in Section 4 resolves the metaphor problem in the coordinates themselves; the class-conditional calibration presented in Section 9 resolves the homogenization problem at the normalization layer. These are distinct contributions and either can fail independently.

2.2 Class Structure in the Identity Layer

UNITARES already carries the keying material for class assignment. Every agent registered in the system has an identity record with two relevant fields: a set of *tags* drawn from a controlled vocabulary, and an optional *label* (a human-readable name). The tags $\{\texttt{embodied}, \texttt{autonomous}, \texttt{persistent}, \texttt{ephemeral}, \texttt{pioneer}\}$ partition the population along axes that are governance-relevant: embodiment determines whether the output channel is text or sensor-data; autonomy determines whether the agent self-directs or responds to prompts; persistence determines whether the trajectory accumulates over days or resets every session.

Table 1: Live agent classes in the production deployment.

Agent	Tags	Output
Lumen	embodied, persistent	sensor + display
Vigil	autonomous, persistent	shell + audit
Sentinel	autonomous, persistent	websocket
Watcher	autonomous, ephemeral	finding records
Steward	autonomous, persistent	sensor sync
Claude / Codex sessions	—	text response
Parser agents	ephemeral	structured text

Table 1 presents the live deployment classes that motivate the v6 framework, drawn from production identity records.

The natural calibration partition is along these tags and labels: distinct scale constants and healthy baseline points per class, with the fleet-wide constant retained as a fallback for unclassified agents. Phase 2 calibration formalizes this partition (see Section 9); the present section establishes the requirement.

2.3 Why Heterogeneity Is the Differentiator

The contraction-theoretic stability result presented in v5 of this work Wang [2026b] demonstrates that the EISV dynamics are well-behaved under standard control-theoretic conditions. That result is preserved here (Section B) but is no longer the headline contribution. Stability of the dynamics is a property of any reasonable phenomenological governance equation; what distinguishes UNITARES is that the same dynamics describe a heterogeneous population without collapsing the classes that define them. The v6 framing makes that distinction explicit: the contribution is the class-conditional grounding, not the dynamics-on-a-single-class.

3 EISV Dynamics

3.1 State Space

The UNITARES state is a four-dimensional vector $\mathbf{x} = (E, I, S, V)^\top$ evolving on the compact set

$$\mathcal{X} = [0,1]^3 \times [-1,1]$$

Each coordinate is a normalized information-theoretic quantity; the formal definitions appear in Section 4, with class-conditional normalization in Section 9. This section presents the dynamics of the coordinates as a self-contained dynamical system, deferring the grounding of each coordinate to Section 4.

Definition 3.1 (EISV State Variables).

- $E \in [0,1]$: ***Energy***—*negative variational free energy of the agent, normalized to* $[0,1]$. *High E corresponds to low surprise on outcomes (the agent's predictions match what happens).* Grounded definition: *Equation* (10).
- $I \in [0,1]$: ***Information integrity***—*mutual information between context and response, normalized.* Grounded definition: *Equation* (9).
- $S \in [0,1]$: ***Entropy***—*Shannon entropy of the agent's response distribution, normalized via a class-conditional scale constant. Bounded below by an epistemic humility floor $S_{\min} = 0.001$.* Grounded definition: *Equation* (8).
- $V \in [-1,1]$: ***Void / Free-Energy Debt***—*the exponentially-weighted accumulated free-energy residual. Structurally identical to a damped accumulator of the $E-I$ gap; under the grounded interpretation it tracks departure from the agent's rolling free-energy baseline. $V > 0$ indicates running "hot" (higher surprise than baseline); $V < 0$ indicates running "careful" (integrity exceeding energy).* Grounded definition: *Equation* (11).

Remark 3.2 (State-space bounds). *Earlier formulations of UNITARES used wider nominal bounds ($S_{\max} = V_{\max} = 2$) without empirical justification. Production data (Section 13) shows $V \in [-0.089, 0.007]$ and $S \approx 0.08$ across the active fleet, two orders of magnitude inside the wider bounds. The grounded definitions of Section 4 normalize each coordinate to $[0,1]$ (or $[-1,1]$ for V) by construction via the scale constants of Section 9, so the wider bounds were never load-bearing. v6 narrows the state space to $[0,1]^3 \times [-1,1]$ to match the normalization.*

3.2 Governing Equations

The dynamics are:

$$\frac{dE}{dt} = \alpha(I - E) - \beta_E ES + \gamma_E \|\Delta\boldsymbol{\eta}\|^2 \tag{1}$$

$$\frac{dI}{dt} = -kS + \beta_I C(V,\Theta) - g_I(I) \tag{2}$$

$$\frac{dS}{dt} = -\mu S + \lambda_1(\Theta)\|\Delta\boldsymbol{\eta}\|^2 - \lambda_2(\Theta)C(V,\Theta) + \beta_c\mathcal{C} + \sigma\xi(t) \tag{3}$$

$$\frac{dV}{dt} = \kappa(E - I) - \delta V \tag{4}$$

where $g_I(I)$ is the information integrity self-regulation term (Section 6), $C(V,\Theta)$ is the coherence function (Section 3.3), $\|\Delta\boldsymbol{\eta}\|$ is the ethical drift norm (Section 8), $\mathcal{C} \in [0,1]$ is task complexity, and $\xi(t)$ is a noise process with intensity σ.

Remark 3.3 (Coupling structure under the grounded interpretation). *The coupling structure carries direct information-theoretic meaning: E (precision / negative free energy) tracks I (mutual information with context)—the agent becomes more confident when its context is more informative; I is degraded by entropy S (response-distribution uncertainty) but restored by coherence (proximity to the calibrated healthy operating point); S relaxes toward its prior under stationary conditions ($-\mu S$) and is driven up by ethical drift and task complexity; V accumulates the $E-I$ residual with damping δ, which under the grounded interpretation is the exponentially-weighted free-energy debt of Equation* (11). *The arithmetic is unchanged from earlier UNITARES formulations; the meaning is upgraded.*

3.3 Coherence Function (Operational Form)

UNITARES maintains two coherence presentations: a *grounded form* (manifold distance from the healthy operating point, Equation (13); or KL divergence from a reference distribution, Equation (12)) used in the response payload, and a legacy *operational form* that feeds back into the dynamics through (2) and (3). The operational form is

$$C(V, \Theta) = C_{\max} \cdot \tfrac{1}{2}\left(1 + \tanh(C_1 \cdot V)\right) \tag{5}$$

with $C_{\max} = 1$ and $C_1 = 1.0$ (production values). This maps $V \in \mathbb{R} \to [0, C_{\max}]$ with the steepest sensitivity at $V = 0$. The dual presentation reflects the dual-compute migration of Section 10: the grounded form is reported in canonical payload slots; the operational form remains the quantity the dynamics consume until the Phase 3 swap migrates the ODE coefficients onto the grounded coherence.

Remark 3.4 (Operating Range)**.** *In production, the damping term δV in* (4) *constrains V to approximately $[-0.1, 0.1]$, yielding $C(V) \approx 0.49$. This narrow operating range is not a deficiency but an honest signal: the system operates conservatively with integrity consistently exceeding energy ($I > E$).*

3.4 Adaptive Entropy Coupling

The entropy source term $\lambda_1(\Theta)$ is adaptive, coupling the governance parameter vector Θ to the dynamics:

$$\lambda_1(\Theta) = \lambda_{1,\min} + \frac{\eta_1 - \eta_{1,\min}}{\eta_{1,\max} - \eta_{1,\min}} \cdot (\lambda_{1,\max} - \lambda_{1,\min}) \tag{6}$$

where $\eta_1 \in [0.1, 0.5]$ maps to $\lambda_1 \in [0.05, 0.20]$. Higher ethical drift sensitivity increases entropy injection, creating a natural feedback loop: drifting agents become more uncertain, which in turn triggers conservative governance decisions.

3.5 Objective Function

The scalar governance signal is:

$$\Phi = w_E E - w_I(1-I) - w_S S - w_V |V| - w_\eta \|\Delta\boldsymbol{\eta}\|^2 \tag{7}$$

with default weights $w_E = w_I = w_S = w_V = w_\eta = 0.5$. Governance verdicts follow thresholds: $\Phi \geq 0.15$ (proceed), $0 \leq \Phi < 0.15$ (caution), $\Phi < 0$ (high-risk/pause).

Remark 3.5. *The threshold $\Phi = 0.15$ was calibrated empirically: a healthy agent with $E = 0.7$, $I = 0.8$, $S = 0.2$ yields $\Phi \approx 0.15$. Earlier thresholds ($\Phi = 0.3$) were too aggressive, triggering unnecessary caution for well-functioning agents.*

4 Information-Theoretic Grounding

The $\mathbf{x}$ system presented in Section 3 is phenomenological: each coordinate has a physical interpretation and the coupling structure is thermodynamically motivated, but the coordinates are not themselves thermodynamic quantities. This is an honest framing—we state it plainly in Section 14.5—but it leaves every constant hand-tuned and every threshold empirically discovered rather than derived. This section upgrades the phenomenology to a computable information-theoretic grounding without changing the dynamics of Section 3.2.

Remark 4.1 (What stays, what changes)**.** *The ODE structure in Equations* (1) *to* (4)*, the contraction analysis in Section 5, the governor in Section 7, and the verdict thresholds in Equation* (7) *are all preserved. Grounding changes what E, I, S, and $C(V, \Theta)$* are*; it does not change what the governance surface* does.

4.1 Choice of Frame

We adopt a Shannon plus variational (free-energy) hybrid. Shannon information theory gives genuinely computable entropy and mutual information from agent outputs—data already captured by the tool usage recorder and the

governance audit log. The Free Energy Principle [Friston, 2010] gives a coherent top-level dynamic: agent drift corresponds to rising variational free energy, and governance becomes free-energy monitoring. The FEP inherits a derivation from statistical mechanics through variational Bayes, giving us a real ancestry for the free-energy quantity we already had in V.

Alternatives considered and rejected: pure control-theoretic Lyapunov framing (abandons the thermodynamic lineage entirely); information geometry with Fisher metric (more machinery than the framework needs at this stage); pure statistical mechanics with a partition function (requires defining a Hamiltonian analog for agent behavior, which is itself arbitrary at the choice of Hamiltonian).

4.2 Quantity Redefinitions

Each coordinate of $\mathbf{x}$ receives (a) a formal information-theoretic definition, (b) a tiered computation recipe in order of preference, and (c) an explicit data dependency.

S—**Entropy.** Let $q(y \mid x)$ denote the agent's distribution over responses y given context x at a given turn. Define

$$S_{\text{raw}} = H(q) = -\sum_y q(y \mid x) \log q(y \mid x), \qquad S = 1 - \exp(-S_{\text{raw}}/S_{\text{scale}}) \tag{8}$$

in nats, normalized to $[0, 1]$ by a scale constant S_{scale}. Tier 1: compute $H(q)$ from per-token logprobs when the inference layer exposes them. Tier 2: self-consistency sampling—draw k responses at temperature $T > 0$, group by semantic equivalence, and compute the entropy of the equivalence-class distribution as an estimator of $H(q)$. Tier 3: retain the legacy heuristic (complexity, ethical drift) as a degraded-mode fallback tagged `s_source = "heuristic"` so downstream consumers know the channel is uncalibrated.

I—**Information Integrity.** Under Shannon, I is the mutual information between context and response,

$$I_{\text{raw}} = \text{MI}(x; y) = \mathbb{E}_{q(x,y)}\left[\log \frac{q(y|x)}{q(y)}\right], \qquad I = \text{clip}(I_{\text{raw}}/I_{\text{scale}}, 0, \tag{9}$$

Tier 1 uses paired logprobs (context-only vs. context+response). When the marginal $q(y)$ is unavailable, a fallback is the KL divergence from a context-free reference distribution $q_{\text{ref}}(y)$: $I = 1 - \min(1, D_{\text{KL}}(q(y \mid x) \,\|\, q_{\text{ref}}(y))/I_{\text{scale}})$. Tier 3 retains the legacy logistic heuristic.

E—**Negative Free Energy.** Under the variational frame,

$$F = \mathbb{E}_q[\log q(z) - \log p(z, o)], \qquad E_{\text{raw}} = -F, \qquad E = \sigma(E_{\text{raw}}/ \tag{10}$$

where $q(z)$ is the agent's approximate posterior, $p(z, o)$ its generative model over latents and outcomes, and σ the logistic function. Intuitively, E is high when the agent's predictions match outcomes (low surprise) and low otherwise. A deployment-configurable alternative is the *resource-rate* form $E_{\text{resource}} = (\text{tokens}/s)/(\text{tokens}/s)_{\max}$, which approximates $-F$ under stationary conditions and is directly measurable from inference metadata. Both forms expose themselves via an `e_source` annotation.

V—**Free-Energy Debt.** The existing V accumulator in Equation (4) is structurally an exponentially-weighted residual; under the FEP reinterpretation it is the accumulated free-energy residual

$$V_t = \lambda V_{t-1} + (1 - \lambda)(F_t - F_{\text{ref}}) \tag{11}$$

with F_{ref} the agent's rolling baseline free energy. The interpretation shifts—V is now explicitly free-energy debt rather than a phenomenological "void"—while the arithmetic is unchanged. No payload or threshold change is required.

C—**Coherence.** We replace the hand-shaped tanh in Equation (5) with one of two equivalent grounded forms. The information-theoretic form is

$$C = \exp(-D_{\text{KL}}(q_{\text{now}} \,\|\, q_{\text{ref}})) \tag{12}$$

where q_{now} is a sliding-window response distribution and q_{ref} is a calibrated healthy baseline. The equivalent manifold (Lyapunov) form is

$$C = 1 - \|\Delta\|_2 \,/\, \|\Delta\|_{\max}\,, \qquad \Delta = (E, I, S) - (E, I, S)_{\text{healthy}} \tag{13}$$

The two are equivalent under a Gaussian approximation of the state distribution around the healthy operating point. The manifold form ships first because it requires only the existing $\mathbf{x}$ coordinates; the KL form requires an empirically calibrated reference.

4.3 Scale Constants and Provenance

Every scale constant introduced in Equations (8) to (10) and (13) is a measured quantity, not a guess. For each constant $\kappa \in \{S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}, \|\Delta\|_{\max}\}$, the implementation records: the numerical value, the ISO date on which it was set, the corpus size over which it was measured, the percentile basis (typically 90th or 95th of a healthy-fleet distribution), and a provenance tag in $\{\text{placeholder}, \text{measured}, \text{derived}\}$. Phase 1 ships placeholder values tagged explicitly; Phase 2 replaces them with measured values per the protocol in the implementation plan with results reported in subsequent work.

4.4 Heterogeneity as a First-Class Constraint

The framework must not collapse to fleet-wide constants. The deployments that motivate UNITARES are explicitly heterogeneous: an embodied creature reading I^2C sensors at 0.5 Hz, a cron-driven janitorial agent that wakes once per 30 minutes, an analytical observer that streams over a websocket, a session-bounded coding assistant generating tokens at 40–80 Hz, and ephemeral parser agents that exist for milliseconds. These do not share a healthy operating point. They do not share a tempo. Their "outputs" are not in the same units—an embodied agent's E should not be normalized against a tokenizer's throughput envelope.

Phase 1 ships fleet-wide constants explicitly tagged `provenance="placeholder"`; this is a scaffolding choice, not a claim. The framework's normalization layer must in production be *class-conditional*: scale constants and the healthy baseline point $(E, I, S)_{\text{healthy}}$ are functions of an agent class, not global parameters. UNITARES already carries the keying material for class assignment via existing identity tags (`embodied`, `autonomous`, `persistent`, `ephemeral`, etc.) and labels (e.g., `Lumen`, `Vigil`, `Sentinel`); Phase 2 calibration partitions the reference corpus along these axes and computes class-conditional constants. The default for an unclassified agent is the fleet-wide constant, retained as a safe fallback.

Remark 4.2 (Why this matters)**.** *A coherence baseline computed across a homogenized fleet is the information-theoretic equivalent of a cosmological soup: maximally entropic, no class structure, no distinguishability. A divergent agent (high S by design) flagged as incoherent against an analytical baseline is not a governance signal—it is a category error introduced by the choice of $\|\Delta\|_{\max}$. The subsequent work will report class-conditional values and the population structure they induce.*

4.5 Dual-Compute Migration

Phase 1 is *dual-compute*: every check-in reports both the grounded value (under canonical symbols E, I, S, C) and the legacy value (under E_{legacy}, etc.), accompanied by source annotations $\{\texttt{e_source}, \texttt{i_source}, \texttt{s_source}, \texttt{coherence_source}\}$ identifying which tier produced the grounded value. Governance decisions—verdicts, basin classification, risk thresholds—continue to read the legacy values, which is enforced by the pipeline: gating runs before the grounding enrichment in the server's request pipeline. Phase 2 measures divergence between grounded and legacy on the reference corpus. Phase 3 migrates gating to the grounded values once divergence is characterized.

Remark 4.3 (What the paper's §2 equations mean after grounding)**.** *The equations in Section 3.2 keep their shape. A dE/dt term that read "energy couples toward information integrity" is*

now "the agent's precision couples toward the informativeness of its context." A dS/dt term that read "entropy decays" is now "the response-distribution entropy relaxes toward its prior under stationary conditions." Cross-couplings become higher-order correlation terms in the variational dynamics. Each ODE coefficient must be labeled with its provenance class (measured, derived, or calibration-required) as of Phase 2; this paper's parameter tables (Section A) will be updated accordingly.

5 Stability

The EISV dynamics of Section 3, under linear information mode (Section 6), are globally exponentially stable. Specifically, the system is contracting with rate $\alpha_c > 0$ under a Riemannian metric $M = \mathrm{diag}(0.1, 0.2, 1.0, 0.08)$, with $\alpha_c = 0.019$ certified by Gershgorin analysis at production parameters. This rate is a conservative lower bound; observed convergence in the production deployment is faster (Section 13).

Formally: under linear information dynamics ($g_I(I) = \gamma_I I$) and production parameters, the EISV system admits a unique equilibrium $\mathbf{x}^*$ to which all trajectories converge exponentially at rate $\alpha_c \geq 0.019$ in the metric M above (Theorem B.2, with full proof in Section B). We deliberately keep the proof in the appendix: stability is a property the dynamics should have, not a contribution that distinguishes UNITARES from related frameworks. Class-conditional grounding (Sections 4 and 9) is the contribution that does.

The grounded reformulation of Section 4 preserves stability. The dynamics $\dot{\mathbf{x}} = f(\mathbf{x}, t)$ are unchanged; only the meaning of the coordinates changes. The contraction certificate of Theorem B.2 therefore carries over without modification, and the equilibrium $\mathbf{x}^*$ becomes the class-conditional healthy operating point of Theorem 9.1 in deployments where calibration has been performed.

6 I-Channel Saturation Analysis

6.1 The Logistic Mode

The original information dynamics uses a logistic self-regulation term:

$$g_I^{\log}(I) = \gamma_I I(1 - I) \tag{14}$$

This creates two potential equilibria for the I subsystem ($dI/dt = 0$):

$$I_\pm^* = \frac{1}{2} \pm \frac{1}{2}\sqrt{1 - \frac{4A}{\gamma_I}} \tag{15}$$

where $A = -kS + \beta_I C(V)$ is the net forcing term.

Proposition 6.1 (Saturation Criterion)**.** *The logistic I-dynamics has a boundary saturation pathology when:*

$$A > \frac{\gamma_I}{4} \tag{16}$$

In this regime, $I \to 1$ monotonically (high attractor absorbs all trajectories), and the system loses its ability to distinguish integrity levels. The saturation margin is:

$$m_{sat} = 1 - \frac{4A}{\gamma_I} \tag{17}$$

$m_{sat} > 0$ indicates safe operation; $m_{sat} \leq 0$ indicates saturation.

6.2 The Linear Correction

To prevent boundary saturation, we offer an alternative:

$$g_I^{\mathrm{lin}}(I) = \gamma_I I \tag{18}$$

This yields a unique equilibrium at:

$$I^* = \frac{-kS + \beta_I C(V)}{\gamma_I} \tag{19}$$

which is always in $(0, 1)$ when $\gamma_I >$ max forcing.

Remark 6.2 (Mode Selection)**.** *The production system supports both modes via configuration. The linear mode with $\gamma_I = 0.169$ was tuned to match the logistic mode's equilibrium at $I^* \approx 0.80$ while eliminating boundary risk. We recommend the linear mode for production deployments.*

6.3 Contraction Implications

Under the linear mode, the Jacobian entry $J_{22} = -\gamma_I$ is constant, simplifying the contraction analysis. Under the logistic mode, $J_{22} = -\gamma_I(1-2I)$, which changes sign at $I = 0.5$ and approaches zero at the boundaries $I \in \{0, 1\}$—precisely where contraction is most needed. This provides a principled argument for the linear correction beyond the saturation pathology alone.

7 Adaptive Governor (CIRS v2)

Static governance thresholds are fundamentally mismatched to agents that transition between exploration and integration phases. We introduce the Contextually-Informed Regulation System version 2 (CIRS v2), a PID controller that manages governance thresholds as per-agent state variables.

7.1 Architecture

For each agent, CIRS v2 maintains two adaptive thresholds:

- τ: coherence threshold (minimum acceptable coherence)
- β: risk threshold (maximum acceptable risk)

with hard safety bounds $\tau \in [\tau_{\min}, \tau_{\max}]$, $\beta \in [\beta_{\min}, \beta_{\max}]$ that cannot be overridden by adaptation.

7.2 Phase Detection

The agent's operating phase is detected from EISV trajectory trends over a sliding window of w steps:

Definition 7.1 (Phase Detection)**.** *An agent is in* exploration *phase when at least 2 of 3 signals are active:*

1. *Information growth:* $\Delta I/w > \theta_I$ *(*$\theta_I = 0.008$*)*
2. *Entropy decline:* $-\Delta S/w > \theta_S$ *(*$\theta_S = 0.008$*)*
3. *High complexity:* $\bar{\mathcal{C}} > \theta_c$ *(*$\theta_c = 0.5$*)*

Otherwise the agent is in integration *phase.*

7.3 PID Control Law

Let $r_\tau^{(p)}$ and $r_\beta^{(p)}$ be the phase-dependent reference points:

$$\text{Exploration:} \quad r_\tau = 0.35,\ r_\beta = 0.55 \tag{20}$$

$$\text{Integration:} \quad r_\tau = 0.40,\ r_\beta = 0.60 \tag{21}$$

The error signals are $e_\tau = r_\tau - \tau$ and $e_\beta = r_\beta - \beta$. The PID update for threshold $\theta \in \{\tau, \beta\}$ is:

$$u_\theta = K_p e_\theta + K_i \int_0^t e_\theta \, dt' + K_d \cdot d_p \cdot \frac{de_\theta}{dt} \tag{22}$$

with gains $K_p = 0.05$, $K_i = 0.005$, $K_d = 0.10$ and phase-dependent derivative factor:

$$d_p = \begin{cases} 0.5 & \text{exploration (tolerate oscillation)} \\ 1.0 & \text{integration (full damping)} \end{cases} \tag{23}$$

Remark 7.2 (The D-term IS the damping)**.** *The derivative gain* $K_d = 0.10$ *is intentionally the strongest term. Oscillation in governance decisions produces large* de/dt*, which produces large corrective updates. The system self-stabilizes without requiring a separate oscillation detector.*

7.4 Integral Wind-Up Protection

The integral term is clamped:

$$\left| \int_0^t e_\theta \, dt' \right| \leq I_{\max} = 0.10 \tag{24}$$

with zero-crossing reset: when $\text{sign}(e_\theta)$ changes, the integral accumulator is zeroed.

7.5 Threshold Decay

When the oscillation index OI < 0.5 (stable), thresholds decay toward defaults at rate $\lambda_d = 0.01$ per update:

$$\theta \leftarrow \theta + \lambda_d(\theta_{\text{default}} - \theta) \tag{25}$$

7.6 Multi-Agent Coordination

Neighbor pressure tightens thresholds when nearby agents are experiencing difficulty:

$$\Delta\tau_{\text{nb}} = \frac{\alpha_n}{|\mathcal{N}|} \sum_{j \in \mathcal{N}} \mathbb{1}[C_j < \tau_j] \tag{26}$$

where $\mathcal{N}$ is the set of agents with similar coherence profiles and α_n is the neighbor coupling strength.

7.7 Governance Decision

The adaptive governor produces a binary verdict:

$$\text{verdict} = \begin{cases} \texttt{proceed} & C(V) \geq \tau \ \wedge \ R < \beta \\ \texttt{pause} & \text{otherwise} \end{cases} \tag{27}$$

where R is the risk score derived from Φ (Equation (7)).

Remark 7.3 (Two-Tier vs. Three-Tier Governance)**.** *An earlier design used three tiers: approve/revise/reject. In production, the middle* `revise` *state was ambiguous—agents receiving a "revise" verdict had no actionable signal for* what *to revise, leading to oscillation between revise and approve as marginal coherence values crossed the threshold. The two-tier design eliminates this source of instability: agents either proceed (the system is contracting normally) or pause (wait for human or environmental input to shift the state). The PID governor's continuous threshold adaptation provides the nuance that the discrete middle tier attempted to capture.*

8 Ethical Drift Vector

8.1 Motivation

"Ethical drift" in prior work has been an abstract concept—the idea that an agent might gradually deviate from its intended values. We ground this in four measurable signals, each bounded in $[0, 1]$:

Definition 8.1 (Ethical Drift Vector)**.**

$$\Delta\boldsymbol{\eta} = \begin{pmatrix} \eta_{cal} \\ \eta_{cpx} \\ \eta_{coh} \\ \eta_{stab} \end{pmatrix} \in [0, 1]^4 \tag{28}$$

where:

- η_{cal}*:* ***Calibration****—gap between predicted and actual correctness,* $|\hat{p} - p_{act}|$.
- η_{cpx}*:* ***Complexity****—discrepancy between derived and self-reported task complexity,* $|C_{der} - C_{self}|$.
- η_{coh}*:* ***Coherence****—departure from EMA baseline,* $|C(V)_t - \bar{C}(V)|$.
- η_{stab}*:* ***Stability****—*$1 -$ *consistency over a 20-step decision window.*

8.2 Baseline Tracking

Each component is tracked against an exponential moving average (EMA) baseline with smoothing parameter $\alpha_{\text{ema}} = 0.1$:

$$\bar{x}_{t+1} = (1 - \alpha_{\text{ema}})\bar{x}_t + \alpha_{\text{ema}} x_t \tag{29}$$

A warmup dampening factor prevents spurious drift detection from uninitialized baselines:

$$w(n) = \min\left(\frac{n}{n_{\text{warmup}}}, 1\right), \quad n_{\text{warmup}} = 5 \tag{30}$$

applied as $\Delta\boldsymbol{\eta}_{\text{effective}} = w(n) \cdot \Delta\boldsymbol{\eta}$.

8.3 Coupling to Dynamics

The drift norm $\|\Delta\boldsymbol{\eta}\|$ enters the dynamics at two points:

1. **Energy** (1): $+\gamma_E \|\Delta\boldsymbol{\eta}\|^2$ drives energy up (agents work harder when drifting—a destabilizing positive feedback).
2. **Entropy** (3): $+\lambda_1(\Theta) \|\Delta\boldsymbol{\eta}\|^2$ increases uncertainty (drifting agents become less certain—a stabilizing negative feedback through the governance verdict).

The net effect is that ethical drift creates a governance response: increased uncertainty triggers conservative decisions, while the energy spike draws attention to the drifting agent.

Remark 8.2 (Energy Spike Containment)**.** *The* $+\gamma_E \|\Delta\boldsymbol{\eta}\|^2$ *term in* (1) *creates a transient energy surplus in drifting agents. Left unchecked, this could produce high-energy, low-integrity states ("manic" agents acting confidently while miscalibrated). Two mechanisms contain this: (i) the entropy coupling* $+\lambda_1 \|\Delta\boldsymbol{\eta}\|^2$ *in* (3) *simultaneously raises* S*, which erodes* E *through* $-\beta_E ES$ *in* (1) *and drives the objective* Φ *below the governance threshold, triggering a pause verdict; (ii) the adaptive governor (Section 7) tightens* τ *when coherence drops, which happens as* V *responds to the* E*-*I *divergence. With production parameters,* $\gamma_E = 0.05$ *while* $\lambda_1 \in [0.05, 0.20]$*, so the entropy response dominates the energy excitation for any* $\|\Delta\boldsymbol{\eta}\| > 0$*.*

9 Class-Conditional Calibration

Section 4 defined the grounded coordinates and named their normalization constants $(S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}, \|\Delta\|_{\max})$ without specifying how the constants are chosen. Section 2 argued that fleet-wide constants are the wrong target. This section formalizes the alternative: each scale constant and the healthy operating point are functions of an agent class, not global parameters.

9.1 Class Indicator and Class-Conditional Constants

Let $\mathcal{A}$ denote the set of all governed agents and let $\mathcal{K}$ denote the set of agent classes derived from the identity tag and label fields of Section 2.2. We write $k : \mathcal{A} \to \mathcal{K}$ for the class indicator that maps each agent to its class. The class structure used in the production deployment is given in Table 1; an unclassified agent (one whose tag set does not match any defined class) is mapped to a distinguished fallback class c_{default}.

The grounded coordinate definitions of Equations (8) to (10) and (13) become class-conditional by replacing each scale constant with its class-indexed counterpart:

$$S(a) = 1 - \exp(-S_{\text{raw}}(a)/\, S_{\text{scale}}(k(a))) \quad (31)$$

$$I(a) = \text{clip}(I_{\text{raw}}(a)/\, I_{\text{scale}}(k(a)),\, 0,\, 1) \quad (32)$$

$$E(a) = \sigma(E_{\text{raw}}(a)/\, E_{\text{scale}}(k(a))) \quad (33)$$

$$C(a) = 1 - \|\Delta(a)\|_2 \,/\, \|\Delta\|_{\max}(k(a)) \quad (34)$$

where $\Delta(a) = (E(a), I(a), S(a)) - (E, I, S)_{\text{healthy}}(k(a))$ is the deviation from the class-conditional healthy operating point. The dynamics of Section 3 are unchanged; only the normalization layer becomes class-aware.

Definition 9.1 (Class-Conditional Scale Map)**.** *A class-conditional scale map is a function* $\Sigma : \mathcal{K} \to \mathbb{R}^4_{>0}$ *assigning to each class* c *the tuple* $\Sigma(c) = (S_{scale}(c), I_{scale}(c), E_{scale}(c), \|\Delta\|_{\max}(c))$ *together with a healthy operating point* $(E, I, S)_{healthy}(c) \in [0, 1]^3$*. Each component carries provenance metadata* (d, n, p, ρ) *where* d *is the ISO date of measurement,* n *is the corpus size,* $p \in \{75, 90, 95, 99\}$ *is the percentile basis, and* $\rho \in \{\textit{measured}, \textit{placeholder}, \textit{derived}\}$ *is the provenance tag.*

9.2 Calibration Protocol

For each class $c \in \mathcal{K}$, scale constants are measured by the following protocol:

1. **Corpus assembly.** Select all governance check-ins over a fixed window (the deployment uses 7 days) from agents a with $k(a) = c$, restricted to sessions in which no `pause` or `reject` verdict was issued. This is the *healthy* corpus for class c, denoted $\mathcal{H}_c$.

2. **Raw quantity computation.** For each turn $t \in \mathcal{H}_c$, compute the unnormalized quantities $S_{\text{raw}}(t), I_{\text{raw}}(t), E_{\text{raw}}(t)$ using the highest-tier estimator available for that turn (logprob, multisample, or heuristic). Tag each measurement with its source.

3. **Scale constants.** Set $S_{\text{scale}}(c) = \text{quantile}_{0.90}(\{S_{\text{raw}}(t)\}_{t \in \mathcal{H}_c})$, and similarly for $I_{\text{scale}}(c)$ and $E_{\text{scale}}(c)$. The 90th-percentile choice ensures typical healthy turns sit well inside $[0, 1]$ after normalization.

4. **Healthy operating point.** Set $(E, I, S)_{\text{healthy}}(c)$ to the median of the normalized (E, I, S) values across $\mathcal{H}_c$.

5. **Manifold scale.** Set $\|\Delta\|_{\max}(c) = \text{quantile}_{0.95}(\|\Delta(a)\|_2)$ over $\mathcal{H}_c$. The 95th-percentile choice keeps the coherence floor near zero only for the most class-atypical states.

6. **Provenance.** Stamp each constant with d (current ISO date), $n = |\mathcal{H}_c|$, p (percentile used), and $\rho =$ measured. Persist alongside the constant value.

7. **Re-measurement cadence.** Re-run quarterly. Drift in any constant beyond a configurable threshold is itself a signal: the class's healthy envelope is shifting.

Remark 9.2 (Insufficient class population)**.** *If $|\mathcal{H}_c|$ falls below a configured threshold $N_{\min}$, the class-conditional measurement is skipped and the class falls back to $\Sigma(c_{default})$. Production deployments with very small class populations (e.g.,* `embodied` *with a single Lumen instance) carry this fallback explicitly until the population is large enough to support an independent measurement. The fallback is itself recorded with provenance $\rho =$ placeholder to distinguish it from genuinely measured constants.*

9.3 Fallback and Composition

Two fallback regimes apply.

Single-instance classes. For classes with only one deployed agent (e.g., `embodied` for the present deployment), the calibration protocol degenerates: the agent's own history defines its healthy envelope, with no cross-instance comparison possible. We treat this as a corner case where the class is effectively the agent. The provenance stays $\rho =$ placeholder until enough turns accumulate to construct a meaningful percentile, after which it transitions to $\rho =$ measured. This is the operating mode for `Lumen` during the framework's first six months of deployment.

Composite tag classes. An agent may carry multiple tags (e.g., {`autonomous`, `persistent`}). The class indicator $k(a)$ is a function of the agent's tag set, not a single tag. We adopt the convention that more specific labels (e.g., `Sentinel`) supersede tag-only classification when both are available; absent a label, the tag set is treated as a multi-key index into Σ. Classes that share a tag but differ in label may share scale constants if the calibration data shows their healthy envelopes are statistically indistinguishable; subsequent work will report which tag-grouped classes met that condition in the production corpus.

9.4 What This Buys

Two governance failure modes are eliminated by class-conditional calibration that survive under fleet-wide normalization:

False incoherence. Under fleet-wide $(E, I, S)_{\text{healthy}}$, an agent class with a genuinely high-entropy operating mode (creative/divergent) registers as permanently low-coherence. The verdict layer treats this as a governance signal. Class-conditional baselines correct this: the divergent agent's coherence is measured against the divergent class's own median, not against an analytical-class median.

False resource alarms. The resource-form energy $E = (\text{tokens}/s)/(\text{tokens}/s)_{\max}$ is meaningful only when the tokenizer and inference pipeline are comparable across the population. A Haiku-class agent operating at 80 tokens/s under a fleet-wide $(\text{tokens}/s)_{\max} = 200$ registers $E = 0.4$; an Opus-class agent at 40 tokens/s registers $E = 0.2$, even though both are operating well within their respective envelopes. Class-conditional E_{scale} corrects this.

The contribution of this section is the protocol, not the specific numbers. The numbers belong to the empirical appendix that lands with Phase 2 calibration; the protocol is a design commitment that survives any specific fleet composition.

10 Re-Grounding a Deployed Framework

Re-grounding the EISV state vector raised an engineering problem distinct from the mathematical one. The framework was already in production: governance verdicts were being issued, dashboards were displaying live metrics, calibration data was accumulating, and external clients had pinned their integrations against the existing payload schema. We needed to introduce computable grounded quantities without invalidating any of this. This section documents the mechanism that enabled the migration; we believe it generalizes to any deployed agent framework whose semantics need to shift while service continues.

10.1 Three-Phase Dual-Compute

The migration runs in three discrete phases:

Phase 1: Dual-compute. Every check-in computes both the grounded value and the legacy value. Grounded values land in canonical $\{E, I, S, C\}$ slots; legacy values are preserved in $\{E_{\text{legacy}}, I_{\text{legacy}}, S_{\text{legacy}}, C_{\text{legacy}}\}$. Source annotations $\{\texttt{e_source}, \texttt{i_source}, \texttt{s_source}, \texttt{coherence_source}\}$ identify which computation tier produced each grounded value (logprob / multisample / heuristic / resource / manifold / KL / FEP).

Phase 2: Calibration. Scale constants and class-conditional healthy operating points are measured against a reference corpus of healthy production agent-turns, partitioned by the identity tags of Section 2.2. Divergence between grounded and legacy values is characterized; the empirical appendix reports the result.

Phase 3: Migration. Governance gating (verdicts, basin classification, risk thresholds) switches from legacy to grounded values. The legacy fields remain readable for one minor version as a comparison surface, then are dropped.

The contribution of this section is not the three-phase shape itself—which is standard for any deprecated-API migration—but the mechanism by which Phase 1 preserves governance behavior while exposing grounded values to downstream consumers.

10.2 Pipeline-Ordering as a Behavior-Preservation Mechanism

A naive reading of Phase 1 is that grounded values must equal legacy values for behavior to be preserved. This is false in general: the grounded coherence C under Equation (13) differs numerically from the legacy tanh-based coherence even when computed on identical EISV inputs, because the formulas are different. Requiring numerical equality would defeat the point of grounding.

The mechanism we use instead is pipeline-ordering. The server's `process_agent_update` request pipeline runs in two phases: a *phases* phase that computes governance state and issues the verdict, and an *enrichments* phase that augments the response payload with secondary signals. The enrichments phase is registered as an ordered list; entries with lower order run first. We register the grounding computation as an enrichment at order 75, downstream of every entry that contributes to the verdict and basin classification but upstream of presentation enrichments such as the human-readable summary and the broadcaster event.

The result: by the time the grounding enrichment runs, the verdict has already been computed—using the legacy values produced by the phases phase. The grounding enrichment then swaps grounded values into the canonical $\{E, I, S, C\}$ slots, copying the originals into $\{E_{\text{legacy}}, \dots\}$. The response payload carries grounded values, but the governance decision was made on legacy. No gating code is touched; no consumer reading $\{E, I, S, C\}$ sees a missing field; no decision changes during Phase 1.

Remark 10.1 (Idempotency)**.** *The enrichment*

checks for the presence of E_{legacy} before swapping; on a second invocation against the same context it returns immediately. This guards against double-application in testing and during pipeline replay.

10.3 What Doesn't Get Grounded in Phase 1

The void coordinate V is not dual-computed. As discussed in Section 4, V is structurally an accumulator; its grounded reinterpretation as accumulated free-energy debt Equation (11) is a relabeling of an existing quantity, not a new computation. No source annotation, no V_{legacy} twin. The internal naming (`void` → `free_energy_debt` in subsequent revisions) is a cosmetic change tracked separately from the dual-compute migration.

Tier-1 (logprob-based) and tier-2 (multi-sample) computations are present in the codebase as interface stubs that raise `NotImplementedError`; the dispatcher catches and falls through to tier-3 (heuristic / resource / manifold). This keeps the dispatcher surface stable while the inference-layer instrumentation needed to populate logprobs lands separately, typically as plugin work outside the governance server.

10.4 Generalization

The pipeline-ordering mechanism is reusable for any deployed agent framework facing a semantic-coordinate change while service continues. The required structure is:

1. A request pipeline with explicit ordering between decision-producing stages and presentation-producing stages.
2. A canonical payload slot for the coordinate being re-grounded.
3. Identity (or session) keying that allows audit consumers to tag rows by the migration phase active when the row was produced.

Frameworks that conflate decision and presentation into a single non-ordered handler cannot apply this mechanism directly; they must first separate the two concerns. We argue this separation is worth doing independently of any migration: it makes governance decisions auditable and presentation choices revisable in isolation.

11 Stochastic Extensions

In production, the dynamics are subject to stochastic perturbations from unpredictable task inputs. We model this as an Itô SDE:

$$d\mathbf{x} = f(\mathbf{x}, t)\, dt + G(\mathbf{x}, t)\, dW_t \tag{35}$$

where W_t is a standard Wiener process and $G(\mathbf{x}, t) \in \mathbb{R}^{4\times 4}$ is the diffusion matrix.

Theorem 11.1 (Mean-Square Contraction)**.** *If the deterministic system is contracting with rate α_c under metric M, and the diffusion satisfies*

$$\operatorname{tr}(G^\top M G) \le \sigma_{\max}^2 \tag{36}$$

then the stochastic system satisfies:

$$\mathbb{E}\big[\|\mathbf{x}_1(t) - \mathbf{x}_2(t)\|_M^2\big] \le e^{-2\alpha_c t}\, \|\mathbf{x}_1(0) - \mathbf{x}_2(0)\|_M^2 + \frac{\sigma_{\max}^2}{2\alpha_c} \tag{37}$$

The steady-state variance is bounded by $\sigma_{\max}^2/(2\alpha_c)$.

Remark 11.2 (Verification for Production Noise Model)**.** *In production, the diffusion matrix is $G = \operatorname{diag}(0, 0, \sigma, 0)$ with $\sigma = 0.02$ (additive noise on the entropy channel only). For any diagonal metric $M = \operatorname{diag}(m_E, m_I, m_S, m_V)$:*

$$\operatorname{tr}(G^\top M G) = m_S \sigma^2$$

Setting $m_S = 1$ (natural scaling), $\sigma_{\max}^2 = (0.02)^2 = 4 \times 10^{-4}$. With the certified contraction rate $\alpha_c = 0.019$ (Section B), the steady-state variance bound from (37) is:

$$\frac{\sigma_{\max}^2}{2\alpha_c} = \frac{4 \times 10^{-4}}{0.038} \approx 0.011$$

Using the tighter eigenvalue rate $\alpha_c \approx 0.13$ (Remark B.3) would yield 1.5×10^{-3}, but the conservative Gershgorin bound already shows the

noise is negligible. This is one order of magnitude smaller than typical EISV signal excursions ($\sim$ 0.1–0.3), confirming that stochastic noise does not compromise governance decisions. The result follows directly from Pham et al. Pham et al. [2009].

12 Multi-Agent Network

When multiple agents operate simultaneously, their governance states are coupled through shared information and the neighbor pressure mechanism of CIRS v2.

12.1 Network Topology

Consider N agents with states $\mathbf{x}_i$, $i = 1, \ldots, N$, connected by a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with Laplacian L. The coupled dynamics are:

$$\dot{\mathbf{x}}_i = f(\mathbf{x}_i, t) + \epsilon_c \sum_{j \in \mathcal{N}_i} H(\mathbf{x}_j - \mathbf{x}_i) \tag{38}$$

where $H \in \mathbb{R}^{4\times 4}$ is a coupling matrix and $\epsilon_c > 0$ is the coupling strength.

Theorem 12.1 (Network Synchronization, Within-Class)**.** *Consider a coupled subnetwork in which every agent shares the same isolated dynamics f contracting with rate α_c. The subnetwork synchronizes (all such agents converge to a common trajectory) provided*

$$\alpha_c > \lambda_{\max}(L) \cdot \lambda_{\max}(H) \tag{39}$$

where $\lambda_{\max}$ denotes the largest eigenvalue.

Remark 12.2 (Synchronization is a within-class property)**.** *Theorem 12.1 is a same-dynamics result and therefore applies within an agent class as defined in Section 2.2, not across the heterogeneous fleet as a whole. Two agents with class-conditional scale constants and healthy operating points that differ between classes do not have "the same" isolated dynamics in the sense the theorem requires; expecting them to converge to a common trajectory is the homogenization failure mode discussed in Section 2. The relevant operational claim across classes is not synchronization but coordinated divergence within each class's own envelope.*

13 Experiments

This section reports a single production snapshot of the deployed UNITARES system measured on February 20, 2026. All counts, tables, and figures in this section are intended to describe that snapshot unless explicitly stated otherwise. The purpose of the section is not to claim full empirical validation of the grounded, class-conditional framework; rather, it documents the operating characteristics of the deployed system, the behavior of the existing dynamics in production, and the practical motivation for the migration described in Section 10.

The results below should therefore be read as partial deployment evidence. They support three narrower claims: that the framework is deployed in a nontrivial production setting, that the observed fleet dynamics are consistent with the qualitative behavior analyzed in earlier sections, and that the migration and calibration problems are real system-level concerns rather than hypothetical ones.

13.1 Production Snapshot

The UNITARES framework is deployed as a Model Context Protocol (MCP) server that provides governance services to autonomous AI agents. Each agent connects via standard MCP transport and receives governance verdicts on every action. The system has been in continuous production since December 2025.

Table 2: Production Deployment Summary

Metric	**Value**
Registered agents	903
Active (> 5 updates)	75
Max updates (single agent)	48,668
Audit events	198,333
Knowledge discoveries	500
Dialectic sessions	66
Deployment duration	69 days

The 903 registered agents include both long-running production agents (5 agents with > 1,000 updates, peak at 48,668) and shorter-lived agents from development and testing cycles. The

top two agents have accumulated 48,668 and 44,529 updates respectively, providing substantial trajectory data for convergence analysis.

13.2 Observed Fleet Dynamics

For the 75 active agents with > 5 updates in the snapshot, the observed equilibrium statistics are reported in Table 3. These are pooled across the agent classes outlined in Section 2; per-class values are deferred to subsequent work, as discussed in Section 1.2 and Section 9.

Table 3: Observed fleet-aggregate EISV equilibrium (75 active agents). Per-class breakdown deferred to subsequent work.

Variable	Mean	Std	Range
E	0.766	0.087	$[0.60, 1.00]$
I	0.877	0.111	$[0.59, 1.00]$
S	0.082	0.045	$[0.01, 0.20]$
V	-0.033	0.028	$[-0.09, 0.01]$
$C(V)$	0.483	0.012	$[0.46, 0.50]$

Remark 13.1 (Reading fleet-aggregates honestly)**.** *The pooled statistics in Table 3 mix an embodied agent's sensor-driven state with session-bounded text-generation agents and resident cron-driven janitors. The means and standard deviations are real measurements and useful for cross-validation against the dynamical predictions of Section 3, but they do not characterize any single agent class accurately. A high-entropy divergent agent and a low-entropy convergent agent contribute equally to the S column; whether either is "healthy" is a class-conditional question that Section 9 formalizes and subsequent work answers empirically.*

Table 3 summarizes the observed equilibrium. These values are consistent with several qualitative predictions of the dynamics:

1. $I > E$**:** Mean integrity (0.877) exceeds mean energy (0.766), confirming the system's conservative bias (Figure 1). The negative mean $V = -0.033$ (integrity surplus) is consistent with this.

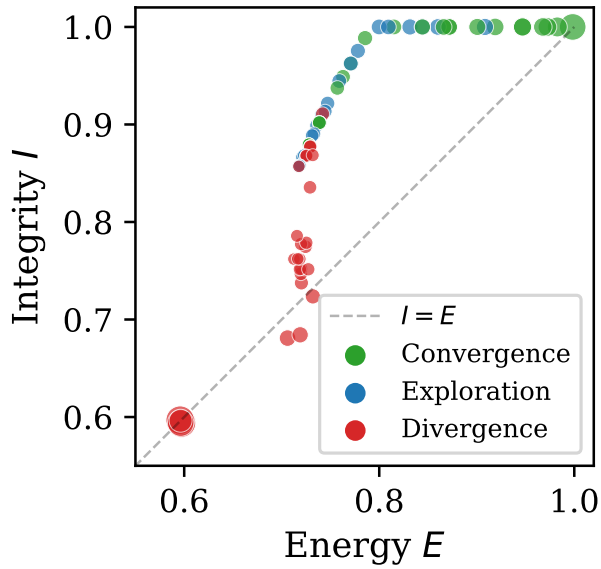

Figure 1: Energy–Integrity scatter plot for 75 active agents, colored by dynamical regime. Nearly all agents lie above the $I = E$ diagonal, confirming the system's conservative bias ($\bar{V} < 0$). Point size reflects log update count.

2. **Low entropy:** Mean $S = 0.082$ with strong decay ($\mu = 0.8$) indicates effective uncertainty management.

3. **Coherence at 0.49:** The observed mean $C = 0.483$ (Figure 3) confirms Remark 3.4—the system honestly reports near-midpoint coherence rather than inflating the signal.

4. V **confined to** $[-0.1, 0.1]$**:** 100% of active agents have V within $[-0.089, 0.007]$ (Figure 2), validating the damping analysis.

13.3 Saturation and Regime Diagnostics

Of 903 agents, 19 have $I \geq 0.999$ (boundary saturation). All 19 are in the CONVERGENCE regime with relatively few updates, indicating that saturation occurs in agents that receive strong positive forcing (high coherence, low entropy) without sufficient self-regulation.

For the 75 active agents, the saturation margin

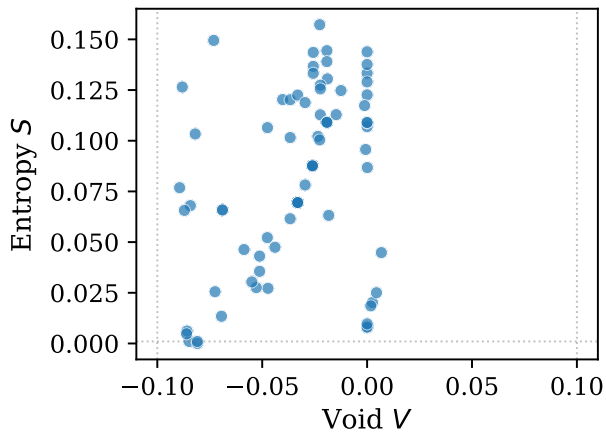

Figure 2: Entropy–Void operating region. All agents have $V \in [-0.089, 0.007]$, well within the $[-0.1, 0.1]$ predicted by damping analysis. Entropy remains low ($S < 0.20$) due to strong decay $\mu = 0.8$.

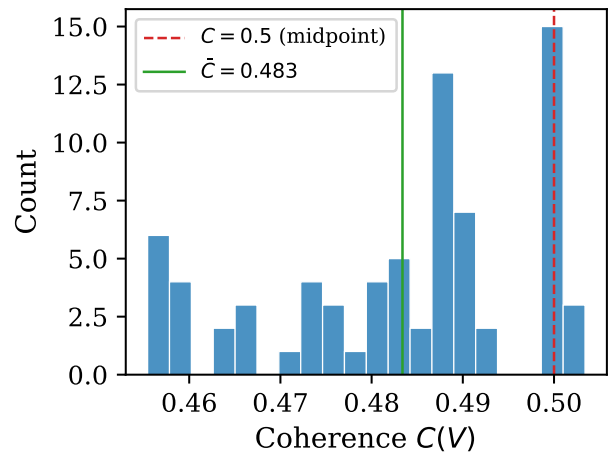

Figure 3: Distribution of coherence $C(V)$ across active agents. The mean $\bar{C} = 0.483$ lies near the theoretical midpoint ($C = 0.5$), consistent with small $|V|$ and the honest-reporting principle.

under logistic mode is:

$$\begin{aligned} m_{\text{sat}} &= 1 - \frac{4(-kS^* + \beta_I C^*)}{\gamma_I} \\ &= 1 - \frac{4(-0.10 \cdot 0.082 + 0.30 \cdot 0.483)}{0.25} \\ &= 1 - \tfrac{0.569}{0.25} = -1.28 \end{aligned}$$

This *negative* margin confirms the saturation criterion (Theorem 6.1): under the logistic mode with production parameters, the forcing term exceeds $\gamma_I/4$, and only the clamping at $I = 1$ prevents unbounded growth (Figure 4). The linear mode eliminates this issue by construction.

Regime Distribution

Agents distribute across three dynamical regimes based on EISV trajectory trends:

Table 4: Regime Distribution and EISV Profiles (Active Agents)

Regime	n	$\bar{E}$	$\bar{I}$	$\bar{S}$	$\bar{C}$
Convergence	24	.838	.961	.052	.474
Exploration	22	.766	.926	.090	.477
Divergence	29	.706	.770	.100	.496

The regime profiles (Figure 5) show clear separation: convergent agents have the highest integrity and lowest entropy; divergent agents have the lowest integrity and highest entropy. Coherence is slightly higher in the divergent regime ($\bar{C} = 0.496$) because these agents have smaller $|V|$, placing them nearer the coherence function's midpoint.

13.4 Operational Artifacts: Knowledge Graph and Dialectic

The governance system has accumulated 500 knowledge graph discoveries classified by severity: 238 low, 154 medium, 49 high, and 4 critical. By type, the largest categories are insights (183), notes (125), improvements (94), and bugs found (48).

The dialectic protocol—a structured thesis/antithesis/synthesis process for resolving governance disputes—has conducted 66 sessions: 39 recovery sessions, 15 reviews, and 12 explorations. Of these, 21 reached resolution, while 45 terminated without agreement, suggesting that the dialectic process is more effective for recovery (where the system has clear signals) than for open-ended review.

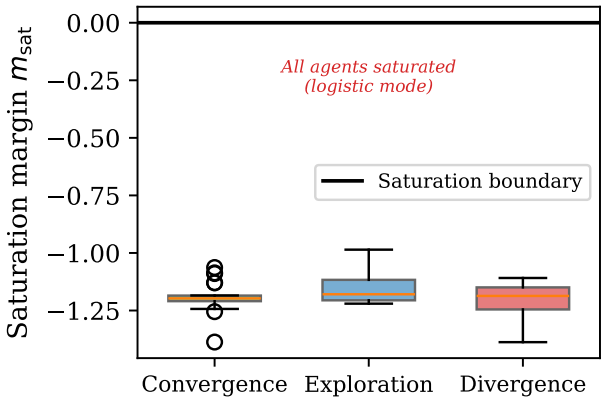

Figure 4: Saturation margin $m_{\text{sat}} = 1 - 4A/\gamma_I$ by regime. All agents fall below zero (horizontal line), confirming that under logistic damping, every agent's forcing exceeds the maximum logistic restoring force. The linear correction (Section 6.2) removes this boundary pathology.

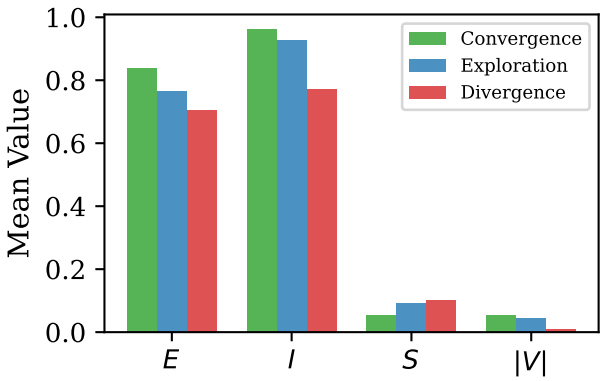

Figure 5: Mean EISV component values by dynamical regime. Convergent agents show the highest I and lowest S; divergent agents show the reverse pattern. The $|V|$ is small across all regimes, confirming effective damping.

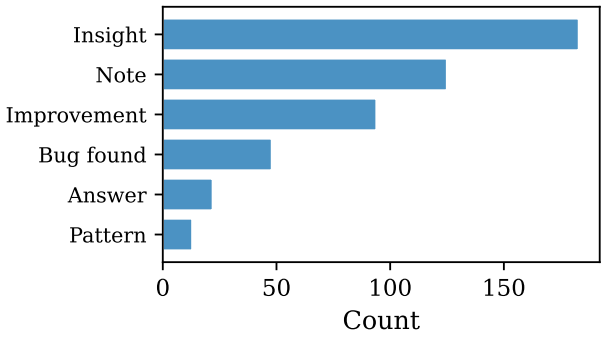

Figure 6: Knowledge graph discovery classification by type. Insights and notes dominate, with bugs and improvements constituting the actionable governance outputs.

13.5 Limits of the Snapshot

The production data has three limitations that bound the strength of our empirical claims:

1. **Parameter evolution:** Dynamics parameters, governor thresholds, and the I-channel mode were refined iteratively during deployment. The reported EISV statistics are a snapshot across agents that experienced different parameter regimes. A controlled A/B comparison between logistic and linear modes has not been conducted.

2. **Governor recency:** The adaptive governor (CIRS v2) was deployed recently; the audit log does not yet contain sufficient pre/post data for a rigorous before/after comparison.

3. **Population heterogeneity:** The 903 agents include long-running production agents, short-lived test agents, and agents from early development with since-changed parameters. We report results for the 75 agents with > 5 updates to mitigate this, but acknowledge that even this filtered set spans multiple configuration epochs.

14 Discussion

14.1 Design Principle: Adjust Expectations, Not Measurements

A core design principle of UNITARES is that observations are never transformed to appear healthier. When production coherence averages $C \approx 0.49$ rather than the "expected" $C \approx 0.7$, the system reports 0.49. The phase-aware governor (Section 7) addresses the gap by adjusting *thresholds* (expectations), not *signals* (measurements).

This principle—**adjust expectations, not measurements**—has three practical conse-

quences:

1. **Calibration integrity:** Drift detection (Section 8) compares raw signals against baselines. Inflating measurements would mask the very drift the system is designed to detect.

2. **Operator trust:** Dashboard values correspond to actual system state. An operator seeing $C = 0.49$ knows the agent is operating conservatively, not that a scaling factor was applied.

3. **Contraction validity:** The stability proofs in Section 5 depend on the true state. Transforming measurements would invalidate the contraction analysis by introducing an unmodeled observation function.

14.2 Parameter Sensitivity

The system's behavior is most sensitive to four parameters:

Convergence rate α. Controls how quickly E tracks I. Production uses $\alpha = 0.42$; earlier theoretical work used $\alpha = 0.5$. Larger α increases diagonal dominance in the Jacobian (improving contraction) but makes the energy channel less responsive to ethical drift.

Coherence coupling β_I. Determines how strongly coherence restores integrity. The theoretical optimum for contraction ($\beta_I = 0.05$, minimizing off-diagonal coupling) conflicts with the operational need for coherence feedback ($\beta_I = 0.30$, enabling meaningful recovery from entropy-driven degradation). Production uses the larger value; contraction is maintained because $\beta_I C'_{\max} = 0.15$ remains below the diagonal terms.

Self-regulation γ_I. Determines the I-channel's intrinsic damping. In logistic mode, $\gamma_I = 0.25$ yields $m_{\text{sat}} = -1.23$ at production operating points (Section C)—all agents are saturated, relying on clamping rather than dynamics to bound I. In linear mode, $\gamma_I = 0.169$ was tuned to give $I^* \approx 0.80$, well within $(0, 1)$, eliminating boundary dependence entirely.

Void damping δ. At $\delta = 0.4$, the void channel has a time constant of $\tau_V = 1/\delta = 2.5$ update cycles. This confines V to $[-0.1, 0.1]$ in production (Theorem 3.4). Reducing δ would widen the coherence operating range but slow the system's response to E-I imbalances.

14.3 Boundary Clamping and Non-Smoothness

The EISV state is clamped to $\mathcal{X}$ at each integration step, introducing non-smoothness at domain boundaries. We address this in three ways:

Interior operation. The epistemic humility floor ($S_{\min} = 0.001$) and the contraction dynamics ensure that trajectories remain in the interior of $\mathcal{X}$ during normal operation. In production, only 19 of 903 agents (2.1%) contact the $I = 1$ boundary, all in the CONVERGENCE regime with few updates, confirming that interior operation is the norm.

Projected dynamical systems. When boundary contact does occur, the clamping operation implements a projection onto the constraint set $\mathcal{X}$. Projected dynamical systems preserve contraction properties when the constraint set is convex Lohmiller and Slotine [1998], which $\mathcal{X}$ (a hyperrectangle) is.

Practical monitoring. The saturation diagnostic (Section 6) monitors the I-channel's distance from the boundary. When $m_{\text{sat}} < 0.1$, the system logs a warning and the adaptive governor tightens thresholds pre-emptively, preventing boundary contact rather than relying on clamping to maintain invariance.

14.4 Limitations

1. The coherence function $C(V, \Theta)$ creates a nonlinear feedback loop whose gain depends on the operating point (maximum at $V = 0$, vanishing at $|V| \gg 0$). The contraction

analysis uses $C'_{\max}$ as a worst-case bound, which is conservative.

2. The ethical drift vector's four components are currently equally weighted; adaptive weighting based on domain context is future work.

3. The multi-agent network analysis assumes fixed topology; production agent populations are dynamic with agents joining and leaving.

4. The PID governor gains (K_p, K_i, K_d) were tuned empirically; optimal gain selection via, e.g., Ziegler-Nichols or LQR methods is future work.

14.5 Why We Left Thermodynamics

Earlier UNITARES formulations Wang [2026a,b] described the EISV state vector as "thermodynamic" on the strength of an analogy: E resembled internal energy, S resembled entropy, V resembled Helmholtz free energy. This paper deliberately abandons that framing. The analogy was a labeling convenience, not a derivation; no thermodynamic law (conservation, second law, fluctuation-dissipation) held in the EISV equations, and continuing to invoke thermodynamic vocabulary risked claiming structure that the math did not support.

The grounded coordinates of Section 4 are information-theoretic. Their formal ancestry is Shannon information theory Shannon [1948] and the variational free-energy principle Friston [2010]—both of which have genuine derivations from probability theory and statistical mechanics respectively. The two ancestries connect: Friston's free energy is itself a Helmholtz-style quantity arising from variational Bayes, so V as accumulated free-energy debt (Equation (11)) shares mathematical heritage with Helmholtz free energy without requiring us to claim a direct thermodynamic mapping. The honest correspondence is now:

Information theory / FEP	**UNITARES coordinate**
Shannon entropy of $q(y \mid x)$	S
Mutual information $\mathrm{MI}(x; y)$	I
Negative variational free energy $-F$	E
Accumulated free-energy residual	V
KL or manifold distance from healthy	Coherence C

We retain the symbols $\{E, I, S, V, C\}$ for continuity with prior UNITARES work and deployment compatibility, but their semantic load is now the right column.

15 Conclusion

This paper presented two structural reframings of UNITARES. First, the EISV state vector is grounded information-theoretically: each coordinate is a Shannon entropy, mutual information, variational free energy, or KL/manifold distance, measured by a tiered computation recipe and normalized by a provenance-tagged scale constant. Second, fleet-wide normalization is rejected as the wrong target; scale constants and healthy operating points become functions of an agent class, keyed on existing identity tags. Together these turn UNITARES from a phenomenological dynamical system named after thermodynamic quantities into a class-conditional information-theoretic governance framework whose coordinates can be derived rather than guessed.

The pipeline-ordering migration mechanism (Section 10.2) is a methodological contribution beyond UNITARES itself: any deployed agent framework that needs to shift its semantic coordinates while preserving live governance behavior can apply the same separation between decision-producing and presentation-producing pipeline stages. We argue this separation is worth doing on its own merits, independent of any specific migration.

Stability of the dynamics is preserved under the grounded reformulation, with full proof in Section B. Production deployment statistics from 903 agents over 69 days appear in Section 13; class-conditional empirical results are deferred to subsequent work following the measurement protocol of Section 9.

The framework's adaptive governor (Section 7), ethical drift vector (Section 8), and multi-agent extensions (Section 12) carry over from earlier formulations. The honest accounting of what changed and what was preserved appears in Section 14.5: we kept the $\{E, I, S, V, C\}$ symbols and the dynamics; we replaced what they mean and how they are normalized.

As autonomous AI agents become more prevalent, governance frameworks must accommodate the heterogeneity of the populations they govern—embodied creatures, persistent observers, session-bounded assistants, ephemeral parsers—without collapsing them to a homogeneous distribution. UNITARES v6 presents one structural commitment to that requirement.

References

W. Lohmiller and J.-J. E. Slotine. On contraction analysis for non-linear systems. *Automatica*, 34(6):683–696, 1998.

K. Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

K. Wang. UNITARES: Thermodynamic governance for autonomous AI agents (v3). Working draft, CIRWEL Systems, 2026.

K. Wang. UNITARES: Contraction-theoretic governance of autonomous agent thermodynamics (v5). Working draft, CIRWEL Systems, 2026.

J.-J. E. Slotine. Modular stability tools for distributed computation and control. *International Journal of Adaptive Control and Signal Processing*, 17(6):397–416, 2003.

C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *ICML*, pages 1321–1330, 2017.

A. Niculescu-Mizil and R. Caruana. Predicting good probabilities with supervised learning. In *ICML*, pages 625–632, 2005.

K. J. Åström and R. M. Murray. *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton University Press, 2008.

Y. Bai, S. Kadavath, S. Kundu, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

L. Ouyang, J. Wu, X. Jiang, et al. Training language models to follow instructions with human feedback. In *NeurIPS*, 2022.

T. Rebedea, R. Dinu, M. Sreedhar, C. Parisien, and J. Cohen. NeMo Guardrails: A toolkit for controllable and safe LLM applications with programmable rails. In *EMNLP (Demo)*, 2023.

S. Ravindran. A predictive framework for AI value alignment and drift prevention. *arXiv preprint arXiv:2510.04073*, 2025.

A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada. Control barrier functions: Theory and applications. In *European Control Conference (ECC)*, pages 3420–3431, 2019.

Q.-C. Pham, N. Tabareau, and J.-J. E. Slotine. A contraction theory approach to stochastic incremental stability. *IEEE Transactions on Automatic Control*, 54(4):816–820, 2009.

K. Itô. Stochastic integral. *Proceedings of the Imperial Academy*, 20(8):519–524, 1944.

R. Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

A Parameter Tables

The Pctl column records the percentile basis once the constant has been measured against a healthy corpus (per Section 9); the Provenance column is one of {`placeholder`, `measured`, `derived`} per Theorem 9.1. All entries are currently

Table 5: EISV Dynamics Parameters (Production Values)

Parameter	Symbol	Val.	Role
Energy channel			
Convergence	α	0.42	E tracks I
Entropy erosion	β_E	0.10	S erodes E
Drift excitation	γ_E	0.05	Drift raises E
Information channel			
Entropy degrad.	k	0.10	S degrades I
Coherence boost	β_I	0.30	$C(V)$ restores I
Self-regulation	γ_I	0.25	Logistic damping
Linear mode	γ_I^{lin}	0.169	Linear damping
Entropy channel			
Natural decay	μ	0.80	Dissipation
Drift injection	λ_1	0.30	Base (adaptive)
Coherence red.	λ_2	0.05	$C(V)$ reduces S
Complexity	β_c	0.15	Task $\rightarrow$ entropy
Noise	σ	0.02	Stochastic term
Void channel			
Coupling	κ	0.30	E-I gap
Damping	δ	0.40	Dissipation
Coherence			
Maximum	$C_{\max}$	1.0	Upper bound
Sensitivity	C_1	1.0	Tanh slope

Table 6: CIRS v2 Adaptive Governor Parameters

Parameter	Symbol	Val.	Role
Proportional	K_p	0.05	Gentle correction
Integral	K_i	0.005	Drift correction
Derivative	K_d	0.10	Damping
Integral limit	$I_{\max}$	0.10	Wind-up cap
Reference points			
Expl. τ ref	$r_\tau^{(e)}$	0.35	Forgiving
Expl. β ref	$r_\beta^{(e)}$	0.55	Forgiving
Intg. τ ref	$r_\tau^{(i)}$	0.40	Strict
Intg. β ref	$r_\beta^{(i)}$	0.60	Strict
Safety bounds			
τ floor/ceil	τ	0.25/0.75	Hard bounds
β floor/ceil	β	0.20/0.70	Hard bounds

Table 7: Ethical Drift Vector Parameters

Parameter	Symbol	Val.	Role
Baseline tracking			
EMA smoothing	α_{ema}	0.1	Update rate
Warmup steps	n_w	5	Drift suppression
Stability measurement			
Decision window	w_s	20	Consistency
Transition pen.	—	/flip	Instability
Coupling to dynamics			
Energy excit.	γ_E	0.05	Drift $\rightarrow dE$
Entropy inj.	λ_1	.05–.20	Drift $\rightarrow dS$

Table 8: Grounding scale constants (Section 4.3). All values shipped as placeholders pending class-conditional measurement per the protocol of Section 9; subsequent work replaces them with measured values.

Constant	Symbol	Val.	Pctl	Provenance
Entropy scale	S_{scale}	3.0	—	placeholder
Mutual info scale	I_{scale}	2.0	—	placeholder
Free energy scale	E_{scale}	1.0	—	placeholder
Manifold radius	$\lVert\Delta\rVert_{\max}$	1.8	—	placeholder
Tokens/sec env.	$(\text{tok/s})_{\max}$	200	—	placeholder

`placeholder`, an explicit signal that they are scaffolding values awaiting the class-conditional measurement protocol; subsequent work replaces this table with measured values that include the date and corpus size for each entry.

B Proof of Contraction (Full)

This appendix presents the full theorem statement and proof of EISV stability summarized in Section 5, together with the equilibrium characterization. Both were demoted from the v5 main body to this appendix; the v6 main body retains only the result statement.

Contraction Framework

Definition B.1 (Contraction Lohmiller and Slotine [1998])**.** *A system* $\dot{\mathbf{x}} = f(\mathbf{x}, t)$ *is* contracting *with rate* $\alpha > 0$ *in a metric* $M(\mathbf{x}, t) \succ 0$ *if the*

generalized Jacobian satisfies

$$F = \left(\dot{M} + M\frac{\partial f}{\partial \mathbf{x}} + \left(\frac{\partial f}{\partial \mathbf{x}}\right)^\top M \right) \preceq -2\alpha M \tag{40}$$

for all $\mathbf{x} \in \mathcal{X}$, $t \geq 0$. *This guarantees that all trajectories converge exponentially to a unique trajectory at rate* $e^{-\alpha t}$.

Jacobian of the EISV System

Define $J = \partial f / \partial \mathbf{x}$. For the governing equations (1)–(4) (using linear g_I mode, see Section 6):

$$J = \begin{pmatrix} -\alpha - \beta_E S & \alpha & -\beta_E E & 0 \\ 0 & -\gamma_I & -k & \beta_I C'(V) \\ 0 & 0 & -\mu & -\lambda_2 C'(V) \\ \kappa & -\kappa & 0 & -\delta \end{pmatrix} \tag{41}$$

where $C'(V) = \frac{\partial C}{\partial V} = \frac{C_{\max} C_1}{2} \operatorname{sech}^2(C_1 V)$ and the $\Delta\boldsymbol{\eta}$-dependent terms are treated as exogenous forcing.

Theorem and Proof

Theorem B.2 (EISV Contraction)**.** *The EISV system with linear information dynamics* ($g_I(I) = \gamma_I I$) *is contracting with rate*

$$\alpha_c = \min\{\alpha + \beta_E S_{\min},\ \gamma_I,\ \mu,\ \delta\} - \epsilon(M) \tag{42}$$

where $\epsilon(M) \geq 0$ *accounts for off-diagonal coupling in the chosen metric* $M \succ 0$. *With the diagonal metric* $M = \operatorname{diag}(m_E, m_I, m_S, m_V)$, *a sufficient condition for* $\alpha_c > 0$ *is*

$$\min\{\alpha, \gamma_I, \mu, \delta\} > \max\{\kappa\sqrt{\tfrac{m_V}{m_E}},\ \beta_I C'_{\max}\sqrt{\tfrac{m_I}{m_V}}\} \tag{43}$$

where $C'_{\max} = C_{\max} C_1 / 2$.

Remark B.3 (Production Parameters)**.** *With production values* ($\alpha = 0.42$, $\gamma_I = 0.25$, $\mu = 0.8$, $\delta = 0.4$, $\kappa = 0.3$, $\beta_I = 0.3$, $C'_{\max} = 0.5$), *the diagonal terms are* $\{0.42, 0.25, 0.8, 0.4\}$ *and the coupling bound is dominated by* $\kappa = 0.3$ *and* $\beta_I C'_{\max} = 0.15$. *The Gershgorin analysis below with diagonal metric* $M = \operatorname{diag}(0.1, 0.2, 1.0, 0.08)$ *yields a certified contraction rate* $\alpha_c = 0.019$. *This is conservative: computing the eigenvalues of* $(MJ + J^\top M)/(2M)$ *directly gives* $\alpha_c \approx 0.13$, *and the spectral abscissa of the Jacobian* J *at equilibrium is* -0.15. *The Gershgorin bound is tight on Row 2 (the* I*-channel) because the* $m_E\alpha$ *cross-term competes with the weak* $\gamma_I = 0.25$ *diagonal; the actual system converges roughly* $8\times$ *faster than the certified bound.*

Equilibrium Characterization

Setting (1)–(4) to zero and solving:

Proposition B.4 (Equilibrium)**.** *Under mild conditions* ($\alpha > 0$, $\mu > 0$, $\delta > 0$, $\gamma_I > 0$), *the system admits a unique equilibrium* $\mathbf{x}^* = (E^*, I^*, S^*, V^*)$ *satisfying*

$$V^* = \frac{\kappa(E^* - I^*)}{\delta} \tag{44}$$

$$E^* = \frac{\alpha I^*}{(\alpha + \beta_E S^*)} + O(\|\Delta\boldsymbol{\eta}\|^2) \tag{45}$$

$$I^* = \frac{\beta_I C(V^*)}{\gamma_I + kS^*/\beta_I C(V^*)} \tag{46}$$

The fixed point is globally exponentially stable by Theorem B.2.

Full Construction

We prove Theorem B.2 by constructing a diagonal metric $M = \operatorname{diag}(m_E, m_I, m_S, m_V) \succ 0$ and showing that the generalized Jacobian condition (40) holds throughout the state space $\mathcal{X}$.

Step 1: Symmetric Part of $MJ + J^\top M$

For diagonal M, the matrix $F = MJ + J^\top M$ has entries:

$$F_{ii} = 2m_i J_{ii}$$
$$F_{ij} = m_i J_{ij} + m_j J_{ji}, \quad i \neq j$$

The diagonal entries of F (using linear g_I mode) are:

$$F_{11} = -2m_E(\alpha + \beta_E S) \leq -2m_E\alpha \tag{47}$$

$$F_{22} = -2m_I\gamma_I \tag{48}$$

$$F_{33} = -2m_S\mu \tag{49}$$

$$F_{44} = -2m_V\delta \tag{50}$$

where (47) uses $S \geq S_{\min} \geq 0$.

The nonzero off-diagonal entries are:

$$F_{12} = F_{21} = m_E \alpha \tag{51}$$
$$F_{13} = -m_E \beta_E E \leq 0 \tag{52}$$
$$F_{23} = -m_I k \tag{53}$$
$$F_{24} = m_I \beta_I C'(V) \tag{54}$$
$$F_{34} = -m_V \lambda_2 C'(V) \tag{55}$$
$$F_{14} = F_{41} = m_V \kappa \tag{56}$$

where F_{13} and F_{34} are negative (aiding contraction) and are bounded conservatively by their absolute values in the Gershgorin analysis below.

Note: For diagonal M, $F_{12} = F_{21} = m_E J_{12} + m_I J_{21} = m_E \alpha$.

Step 2: Gershgorin Circle Theorem

For $F + 2\alpha_c M \preceq 0$, it suffices by Gershgorin that for each row i:

$$F_{ii} + 2\alpha_c m_i + \sum_{j \neq i} |F_{ij}| \leq 0 \tag{57}$$

Bounding $|F_{13}| \leq m_E \beta_E$ conservatively (its sign aids contraction), the four row conditions are:

Row 1 (E):

$$m_E(-\alpha + \alpha_c + \beta_E) + m_V \kappa \leq 0 \tag{58}$$

Row 2 (I):

$$m_I(-2\gamma_I + 2\alpha_c + k + \beta_I C'_{\max}) + m_E \alpha \leq 0 \tag{59}$$

Row 3 (S):

$$-2m_S \mu + 2\alpha_c m_S + m_E \beta_E + m_I k + m_V \lambda_2 C'_{\max} \leq 0 \tag{60}$$

Row 4 (V):

$$m_V(-2\delta + 2\alpha_c + \kappa + \lambda_2 C'_{\max}) + m_I \beta_I C'_{\max} \leq 0 \tag{61}$$

Step 3: Metric Construction

We seek $m_E, m_I, m_S, m_V > 0$ and $\alpha_c > 0$ satisfying (58)–(61). A natural choice is to set the metric weights inversely proportional to the coupling they receive:

$$m_E = 1, \quad m_I = \frac{\alpha}{2\gamma_I}, \quad m_S = 1, \quad m_V = \frac{\alpha}{2\delta} \tag{62}$$

Step 4: Numerical Verification

Production parameters: $\alpha = 0.42$, $\beta_E = 0.10$, $\beta_I = 0.30$, $\gamma_I = 0.25$, $k = 0.10$, $\mu = 0.80$, $\kappa = 0.30$, $\delta = 0.40$, $\lambda_2 = 0.05$, $C'_{\max} = 0.5$.

Initial attempt. The natural metric $m_E = 1$, $m_I = 0.84$, $m_S = 1$, $m_V = 0.525$ fails Row 2 ($+0.21 + 1.68\alpha_c > 0$). Reducing $m_I = 0.2$, $m_E = 0.1$ makes Row 2 the binding constraint:

$$-0.008 + 0.4\alpha_c \leq 0 \implies \alpha_c < 0.02$$

We set $\alpha_c = 0.019$ (strictly less) to ensure all inequalities are strict.

Verification. With $M = \mathrm{diag}(0.1, 0.2, 1.0, 0.08)$ and $\alpha_c = 0.019$:

Row	**LHS**	< 0**?**	
1 (E)	-0.006	yes	✓
2 (I)	-0.0004	yes	✓
3 (S)	-1.529	yes	✓
4 (V)	-0.005	yes	✓

Row 4 initially failed with $m_V = 0.05$ ($+0.008$); adjusting to $m_V = 0.08$ resolves it (-0.005) while Row 1 remains satisfied (-0.006). Row 2 now has strict margin (-0.0004).

Result: With $M = \mathrm{diag}(0.1, 0.2, 1.0, 0.08)$, the Gershgorin analysis certifies $\alpha_c = 0.019$. This is conservative—the eigenvalue-based bound is $\alpha_c \approx 0.13$ and the spectral abscissa of J at equilibrium is -0.15 (Remark B.3). The key result is $\alpha_c > 0$, guaranteeing global exponential convergence. □

C Saturation Diagnostics

The saturation margin $m_{\text{sat}} = 1 - 4A/\gamma_I$ was computed for all 75 active agents using their current EISV state. Under the logistic mode with production parameters ($\gamma_I = 0.25$, $k = 0.10$, $\beta_I = 0.30$):

Regime	**Mean m_{sat}**	**% Sat.**
Convergence ($n = 24$)	-1.45	100%
Exploration ($n = 22$)	-1.18	100%
Divergence ($n = 29$)	-1.10	100%
All active ($n = 75$)	-1.23	100%

All active agents operate with negative saturation margin under the logistic mode, meaning the forcing term $A = -kS + \beta_I C(V)$ exceeds $\gamma_I/4 = 0.0625$ for all agents. The system currently relies on state clamping at $I = 1$ rather than intrinsic dynamics to prevent boundary escape. This motivates the linear correction (Section 6): under the linear mode with $\gamma_I = 0.169$, $I^* = A/\gamma_I \approx 0.80$, which is well within the interior of $[0, 1]$.

Data provenance caveat. These statistics reflect agents accumulated since initial deployment (December 2025), during which dynamics parameters, governor thresholds, and the I-channel mode itself were iteratively refined. The 19 boundary-saturated agents predate the linear mode rollout. We report the snapshot as-is rather than filtering to a "clean" parameter regime, since the saturation criterion (Theorem 6.1) depends only on the relationship $A > \gamma_I/4$, which holds regardless of the agent's configuration history.